NBP Gas & European Power — a focused trading + advisory system

> A focused system that **really trades and advises** the UK NBP natural-gas hub and the European power markets. Built on the ATLAS + envelope core, but with a single, sharp edge: **weather, storage, the merit order and the spark spread**.

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1. What this system does

One market complex, two commodities, one relationship:

- **NBP gas** — the UK gas hub (p/th, GBP), the marginal fuel for UK power.
- **European power** — day-ahead and forward electricity, set by the merit order.
- **The link** — gas is the marginal fuel, so gas price and carbon drive power price through the **spark spread**.

The system **advises** (publishes a bullish/bearish view with a reason) and **trades** (when the view is a tier-1 signal).

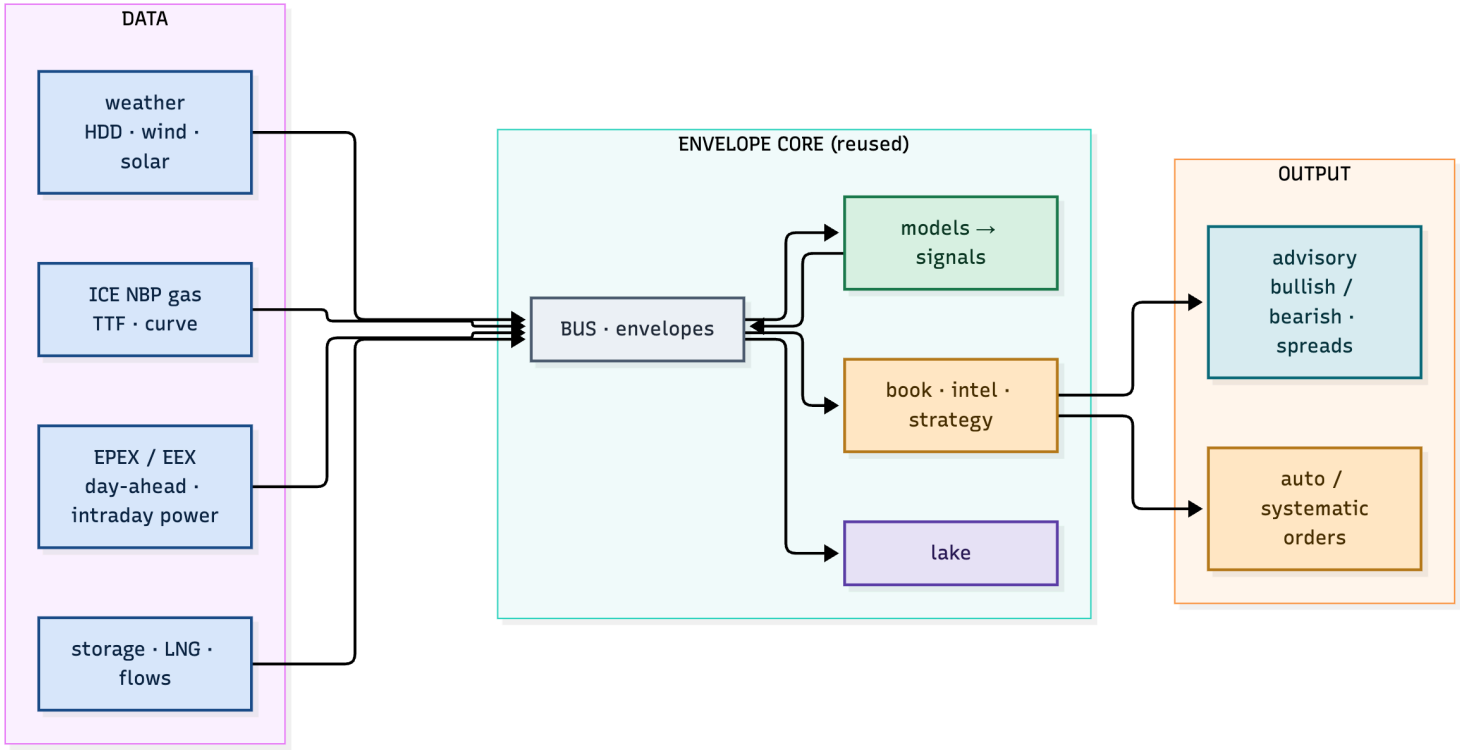


Figure 1.1 — data in, models on the envelope core, advisory + auto out.

2. The NBP gas balance

NBP price is a balance: supply against weather-driven demand, buffered by storage.

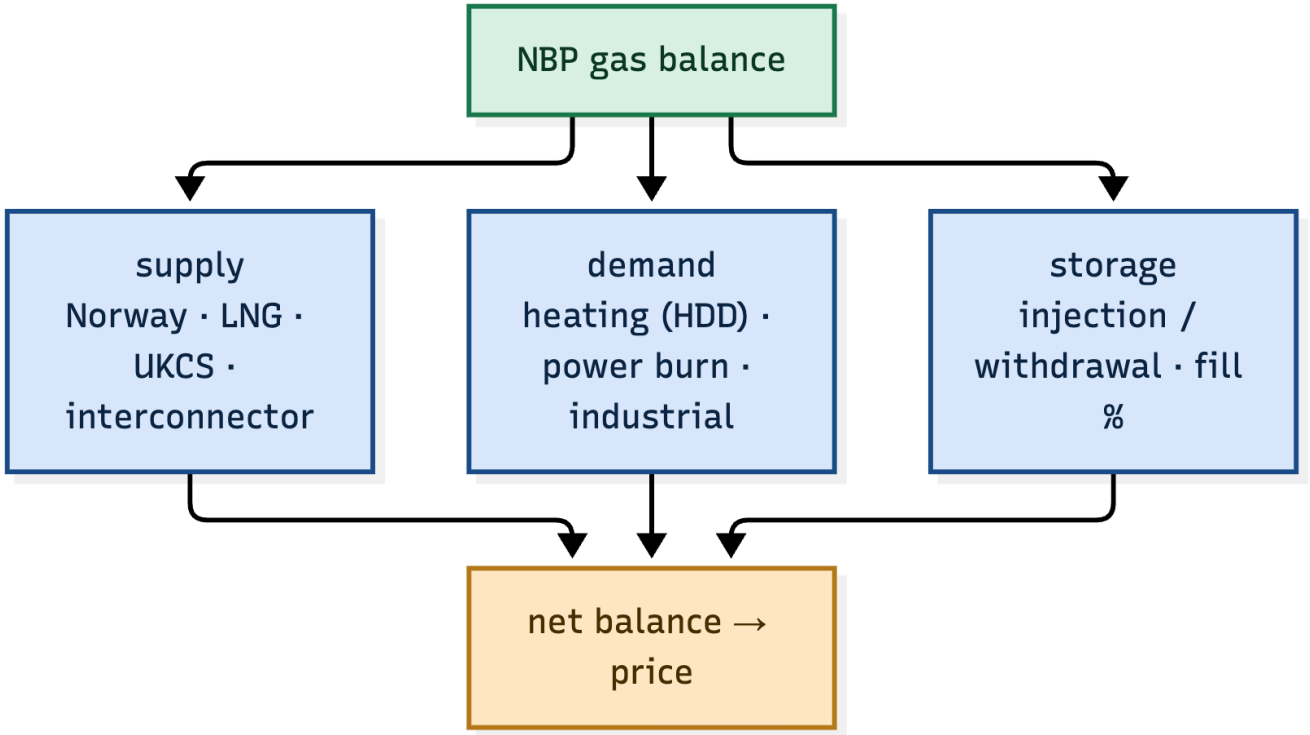


Figure 2.1 — Norway, LNG, UKCS and interconnectors supply; heating and power burn demand.

Component	What drives it	Signal
supply	Norway flows, LNG arrivals, UKCS	more supply → bearish
demand	heating degree days, power burn	cold → bullish
storage	injection vs withdrawal, fill %	low storage → bullish into winter

The balance is the model: **tight balance (low storage + cold) = bullish NBP; loose balance (warm + high storage) = bearish.**

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3. The power merit order

Power price is set by the **marginal plant** — the most expensive unit needed to meet demand. Renewables come first (zero cost), then nuclear, then coal, then gas.

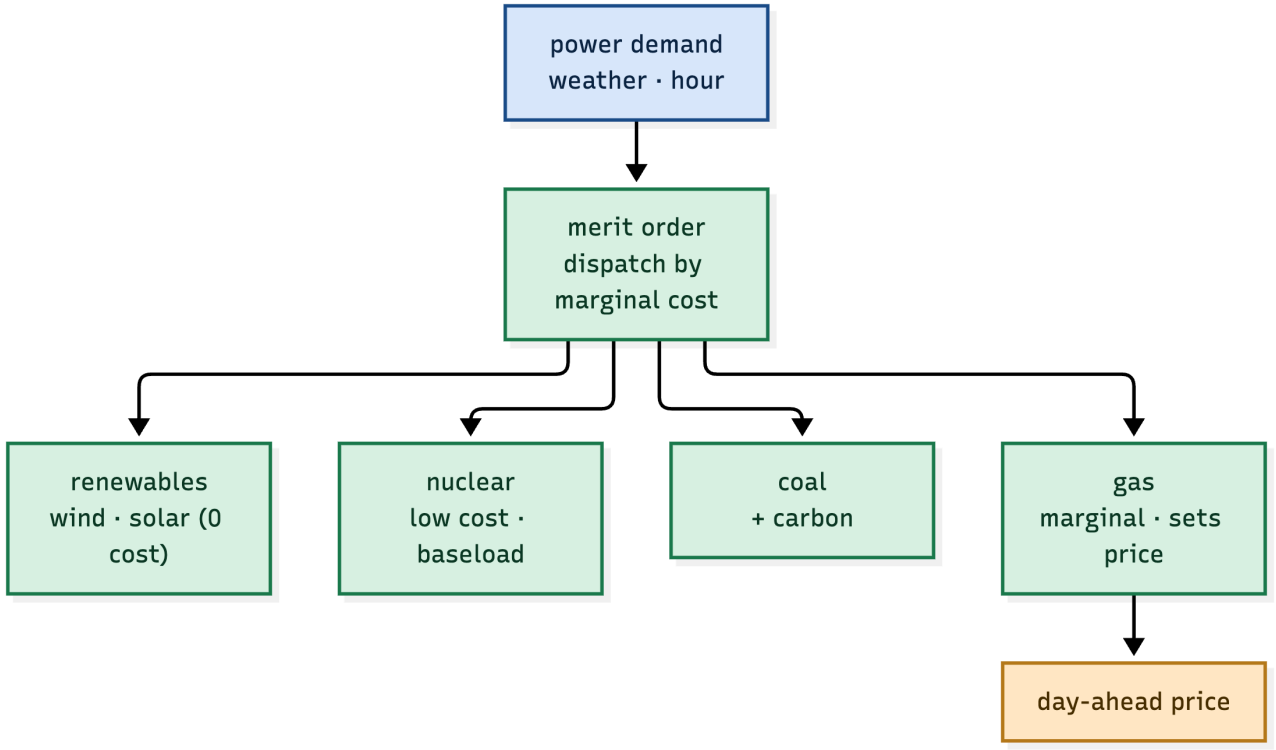


Figure 3.1 — gas is usually the marginal plant, so gas + carbon set the power price.

Stack	Marginal cost	Effect on price
wind / solar	~0	more renewables → lower price
nuclear	low	baseload
coal	mid + carbon	higher
gas	high	**sets the clearing price**

The insight: **gas is usually the price-setter**, so the power price is gas × heat-rate + carbon × intensity + a margin. That is the spark spread.

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4. The spark spread — the one relationship that matters

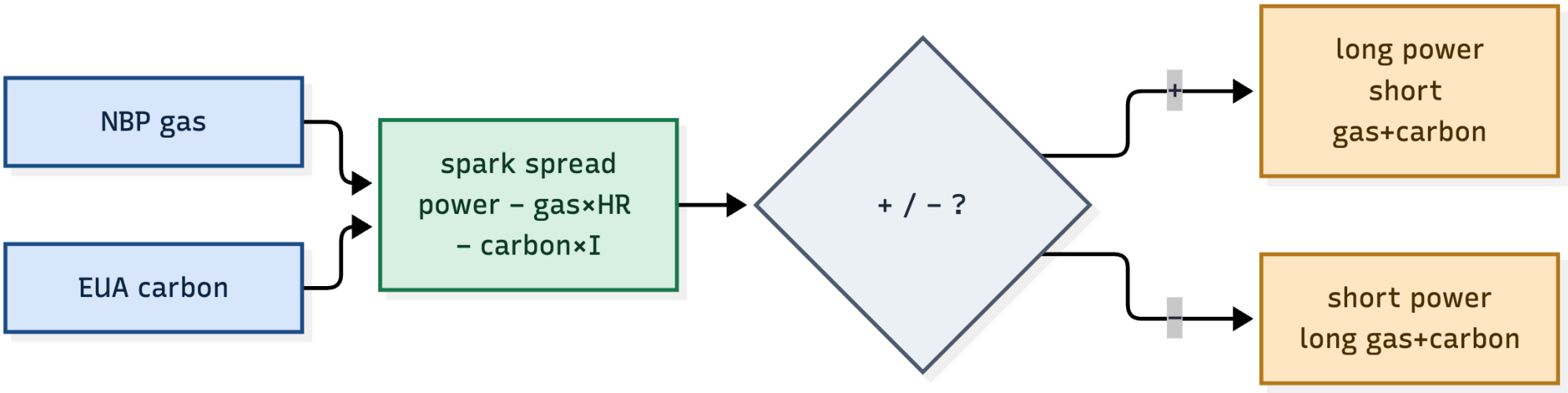


Figure 4.1 — power minus gas and carbon, the clean margin of a gas plant.

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spark spread = power price − (gas price × heat-rate) − (carbon × emission intensity)
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Condition	Trade
spark spread high (plant is profitable)	long power, short gas + carbon
spark spread low / negative (plant losing)	short power, long gas + carbon

The spark spread is the **fundamental anchor** of the whole complex: it ties the two commodities together and gives the system a mean-reverting relationship to trade.

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5. Weather — the driver of both

Weather is the single biggest short-term driver of gas demand and renewable power supply.

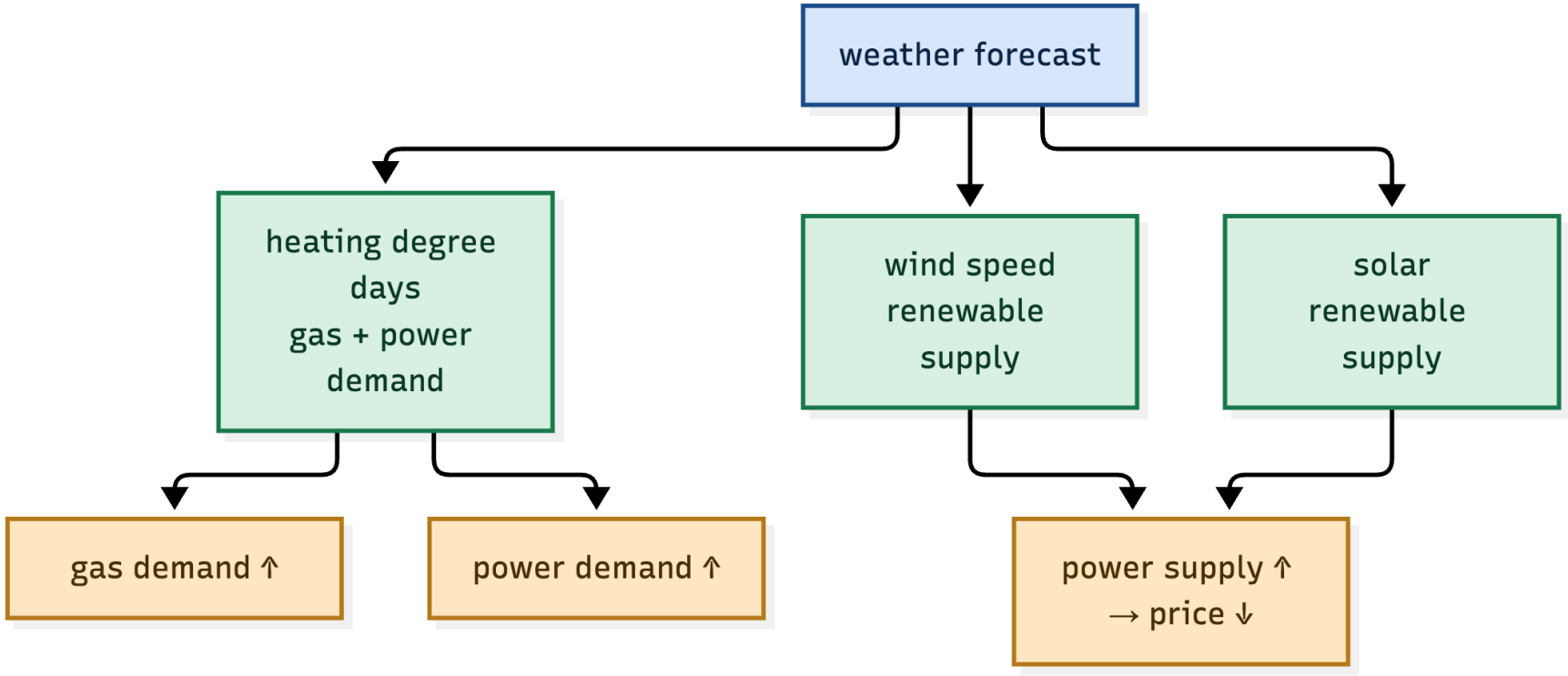


Figure 5.1 — cold raises gas + power demand; wind and solar raise power supply.

Weather	Effect
cold (high HDD)	more heating → more gas + power demand → bullish both
windy / sunny	more renewables → more power supply → bearish power
calm / cloudy	less renewables → gas sets power → bullish power

The weather forecast is a **first-class envelope producer**: it publishes HDD, wind and solar forecasts, and the models consume them.

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6. The products

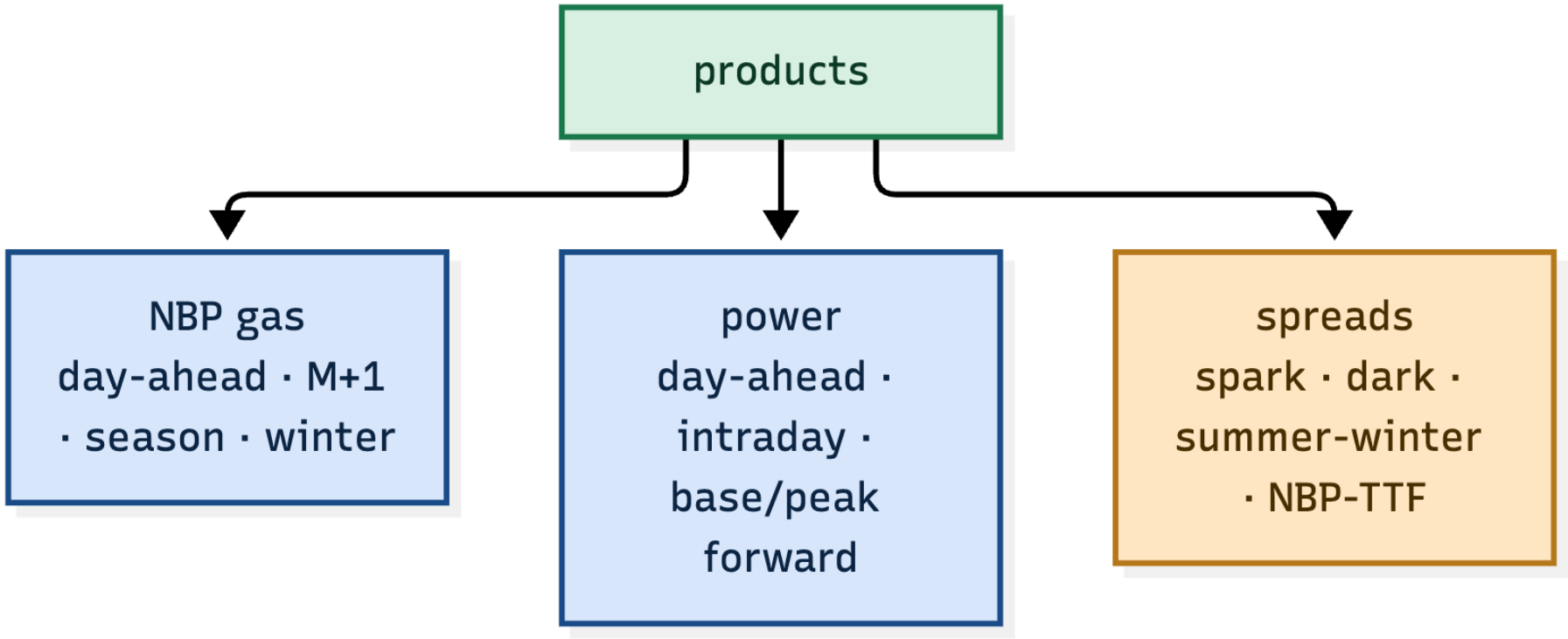


Figure 6.1 — gas, power and the spreads that link them.

Product	Example
NBP gas	day-ahead, month-ahead (M+1), season, winter
power	day-ahead, intraday, base/peak forwards
spreads	spark, dark, summer-winter, NBP-TTF

The system trades the **spreads** first — they are the mean-reverting relationships with a fundamental anchor — and the outrights only when the balance gives a strong directional view.

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7. The advisory — what it actually says

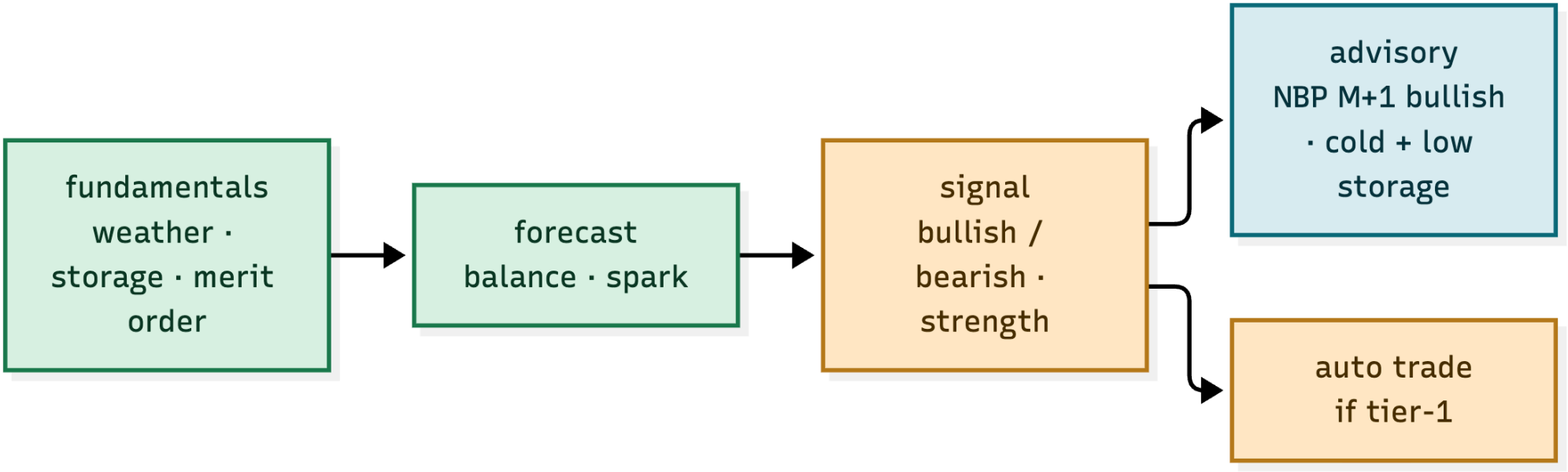


Figure 7.1 — the same signal drives a human-readable advisory and, at tier-1, an auto trade.

Signal	Advisory (verbatim)
cold + low storage + tight LNG	"NBP M+1 bullish — cold snap, storage below 5yr avg"
high wind forecast + mild	"power day-ahead bearish — wind up, demand soft"
spark spread stretched high	"spark rich — long power, short gas+carbon"

The advisory is not a score — it is a **sentence with a reason**, because a trader (or a compliance reviewer) needs the *why*, not just the number.

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8. Data sources

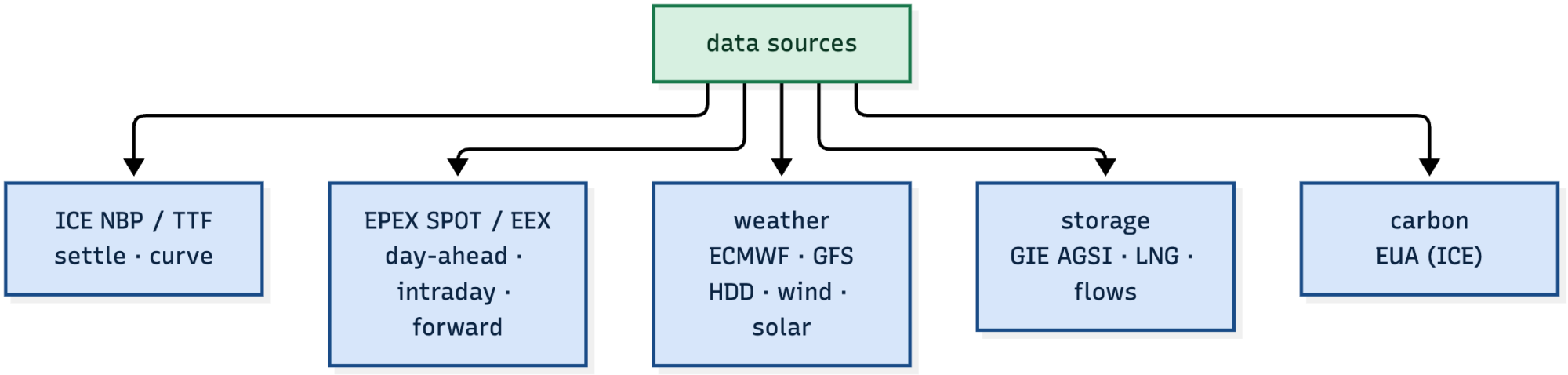


Figure 8.1 — ICE, EPEX/EEX, weather, storage and carbon feed the models.

Source	What it provides
ICE	NBP / TTF gas, EUA carbon — settle and forward curve
EPEX SPOT / EEX	day-ahead and intraday power, forward power
ECMWF / GFS	weather — HDD, wind, solar forecast
GIE AGSI	European gas storage fill levels
ENTSO-E	generation mix, cross-border flows

9. Build plan — the focused version

Phase	What	Deliverable
1	data ingestion (ICE, EPEX, weather, storage)	envelopes on the bus
2	gas balance + merit order models	fundamental forecast
3	spark-spread model	the anchor relationship
4	advisory + tier signals	human-readable views + auto
5	execution (ICE / EEX via FIX)	orders + fills on the bus
6	margin + risk + compliance	the governor + reports

10. Why this system can really trade

- **It has an anchor.** The spark spread is a real, mean-reverting relationship, not a technical pattern — it can be valued and it has a reason to revert.
- **It has a driver.** Weather and storage are measurable, forecastable, and move price on a known timescale.
- **It has a reason for every signal.** The advisory carries the *why*, so a trade is explainable and auditable — which is the envelope's whole promise.

That is a system that can really trade, and really advise.